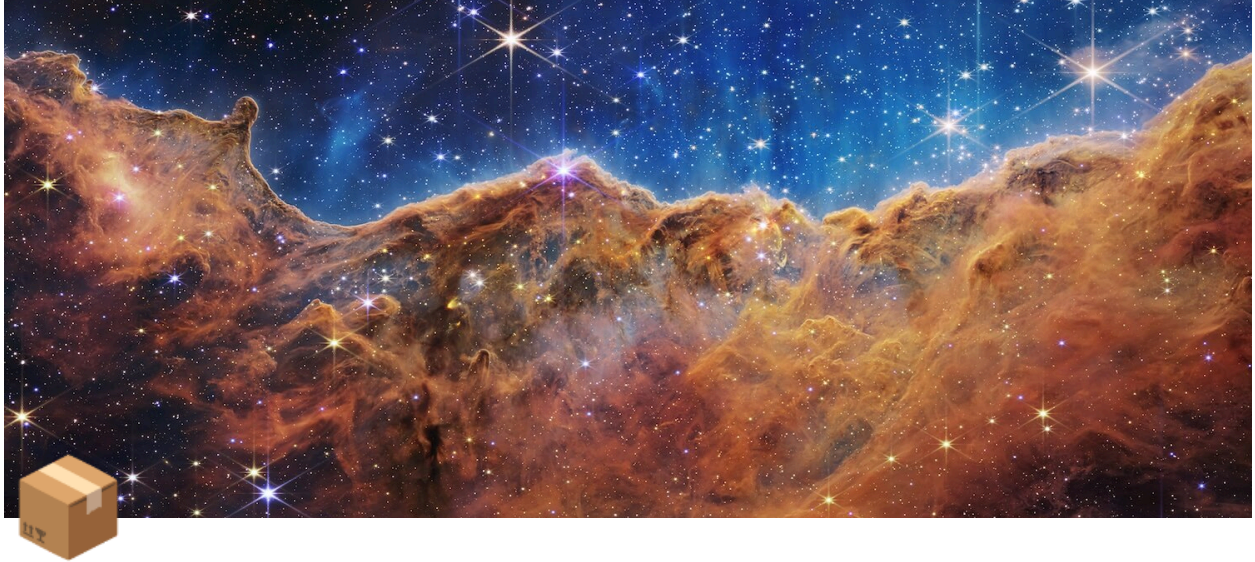




CML6032 ASSIGNMENT 1 — FILES FOR DOWNLOAD



EVERY DELIVERABLE FOR CML6032 ASSIGNMENT 1, PARKED HERE SO IT CAN BE DOWNLOADED WITHOUT REBUILDING ANYTHING. THIS IS A MIRROR OF github.com/0nmeesheal/CML6032-Assignment1, TAKEN AFTER THE VIDEO WAS APPROVED.

THE HAND-INS

SUBMISSION SHEET — ONE PAGE, AS THE BRIEF ASKS

[attachment:c87ff485-1146-4090-aa5c-417a65cbe8bf:Nisheal_2025CYS709_0_Assignment1.docx](#)

SLIDE DECK — 18 SLIDES, 15 NUMBERED ANSWERS PLUS 3 QUESTION SLIDES

[attachment:1b85a83e-fbb1-42a9-a505-db159d185f5a:CML6032_Assignment1_Nisheal_2025CYS7090.pptx](#)

WALKTHROUGH — 22:26, NARRATED IN MY OWN VOICE

[attachment:ea64626f-7e65-44ea-9347-ae2f7b3437af:CML6032_Assignment1_Nisheal_2025CYS7090.mp4](#)

THE WHOLE REPOSITORY — 85 FILES, 61 MB ZIPPED

[CML6032-Assignment1_repo.zip](#)

WHAT IS INSIDE THE ZIP

PATH	WHAT
Nisheal_2025CYS7090_Assignment1.docx	THE SINGLE-PAGE SUBMISSION SHEET
slides/	THE 18-SLIDE PPTX, THE 18 PNG RASTERS AND THE PYTHON RENDER SOURCE
cif_prototypes/	NINE SINGLE-PHASE P1 CIFS, ONE COMPOUND EACH, WITH OXIDATION STATES
cif_batio3/	NINE BaTiO ₃ STATES, Ti FRACTIONAL z FROM 0.480 TO 0.520
renders/	NINE BALL-AND-STICK PNG PLUS FIVE ANNOTATED BaTiO ₃ FRAMES
figures/	THE TWO-BRANCH HYSTERESIS LOOP, DARK AND LIGHT, PNG AND PDF
docs/	THE GITHUB PAGES PLAYER, THE MP4 AND chapters.json

PATH	WHAT
<code>video/</code>	<code>BUILD_VIDEO.py</code> , THE COLAB NOTEBOOK, <code>timeline.csv</code> AND THE CHANGE LOG
<code>README.md</code>	METHOD, NOTATION, SOURCES AND CHECKS
<code>build_docx.py</code> , <code>patch_cif_charges.py</code>	REGENERATE THE SHEET AND THE CIF CHARGE BLOCKS

NOTATION USED THROUGHOUT

LENGTHS IN pm, NEVER ANGSTROM, EXCEPT WHERE THE ISSUED BRIEF IS QUOTED VERBATIM ON THE THREE QUESTION SLIDES. RATIOS AS RATIOS, NEVER PER 100. OXIDATION STATES AS SINGLE TUPLES. MIRROR PLANES AS MP. ROTOINVERSION AXES AND SPACE GROUPS WITH A TRUE OVERHEAD BAR: $m\bar{3}m$, $Fm\bar{3}m$, $Pm\bar{3}m$, $F43m$, $6\bar{m}2$, $\bar{3}m$. THE ONE EXCEPTION IS THE COMMENT HEADERS INSIDE THE CIF FILES, WHICH STAY IN ASCII HYPHEN FORM BECAUSE A NON-ASCII BYTE CAN STOP VESTA OPENING A FILE.

STILL OUTSTANDING



THREE THINGS ARE NOT DONE, AND TWO OF THEM NEED ME RATHER THAN THE BUILD.

1. THE TWO SPECIMEN IDENTITIES AS ISSUED IN CLASS, AND THE MEASURED $\{100\}/\{111\}$ INTERFACIAL ANGLE. BOTH ARE FLAGGED IN RED ON SLIDE 15 AND IN THE DOCX.
2. GITHUB PAGES IS NOT ENABLED YET: SETTINGS, THEN PAGES, THEN BRANCH `main` AND FOLDER `/docs` . THE REPOSITORY MUST BE PUBLIC, AND THE LIVE URL SHOULD BE TESTED WHILE LOGGED OUT.
3. THE SLIDE 15 NARRATION TAKE IS 4.4 s AGAINST A MEDIAN OF 96.4 s, SO THE VERIFICATION AND MANIFEST SLIDE IS EFFECTIVELY UNNARRATED. RE-RECORD IT AND RERUN `BUILD_VIDEO.py` ; ONLY THAT TAKE IS REPROCESSED.